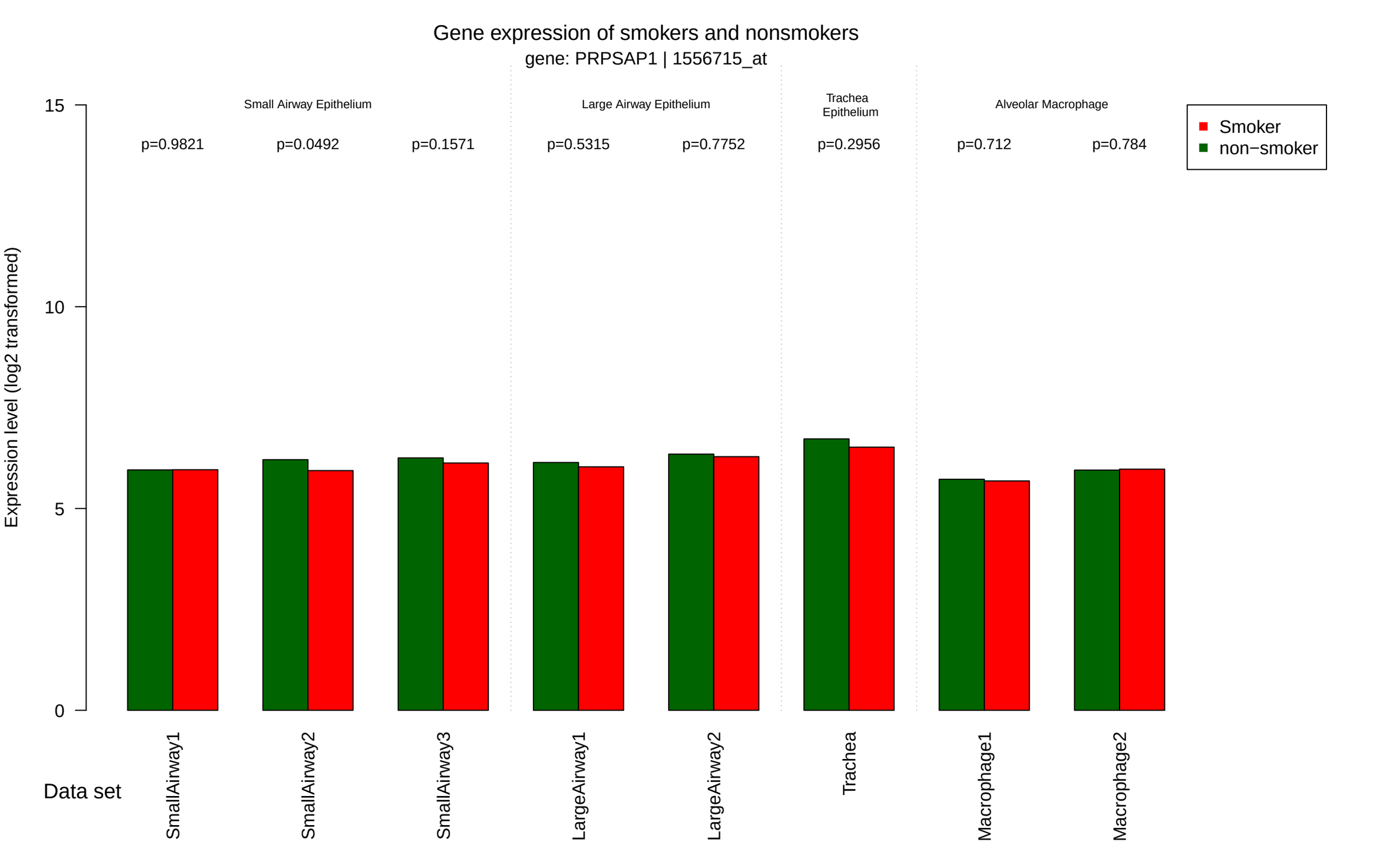
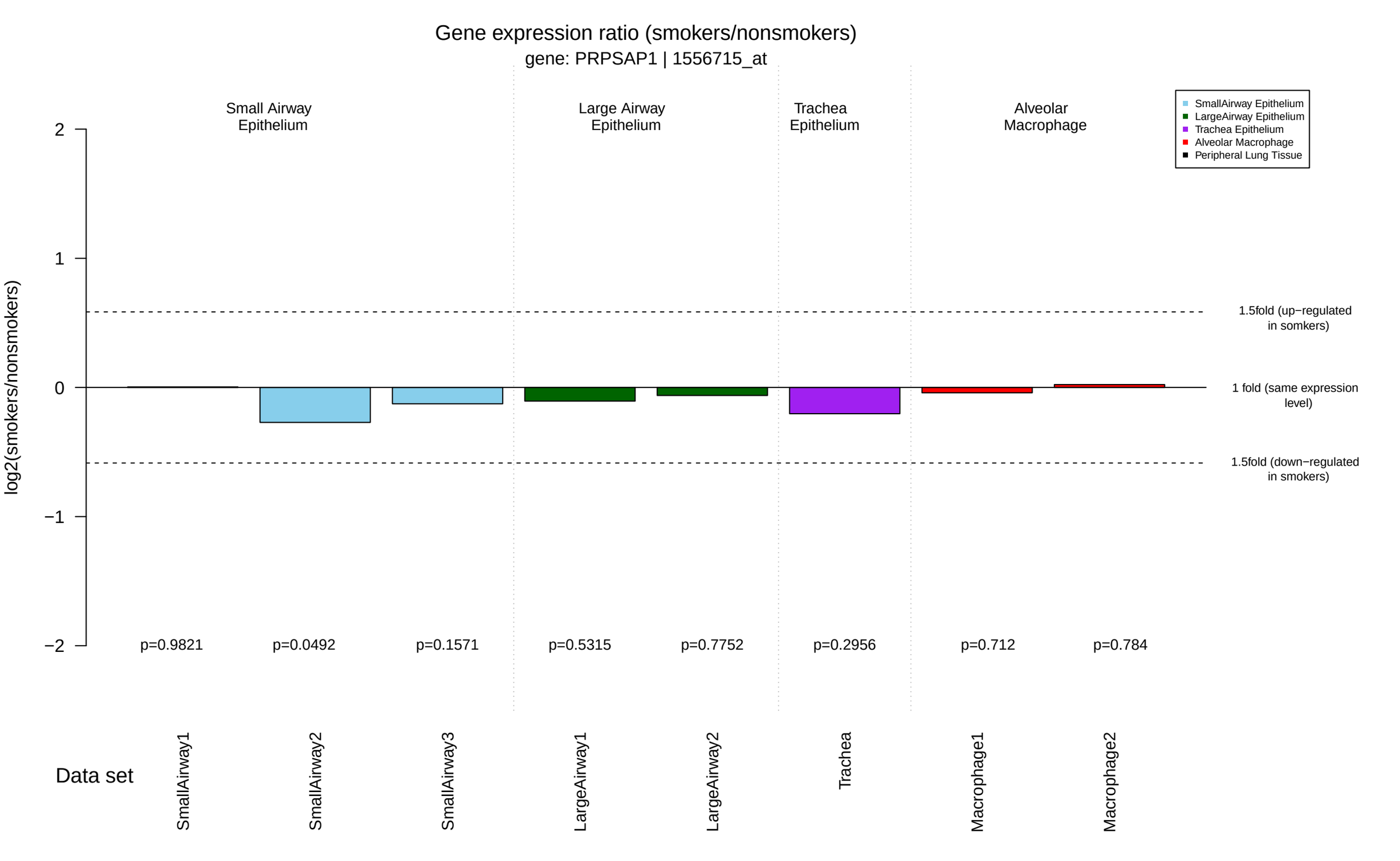
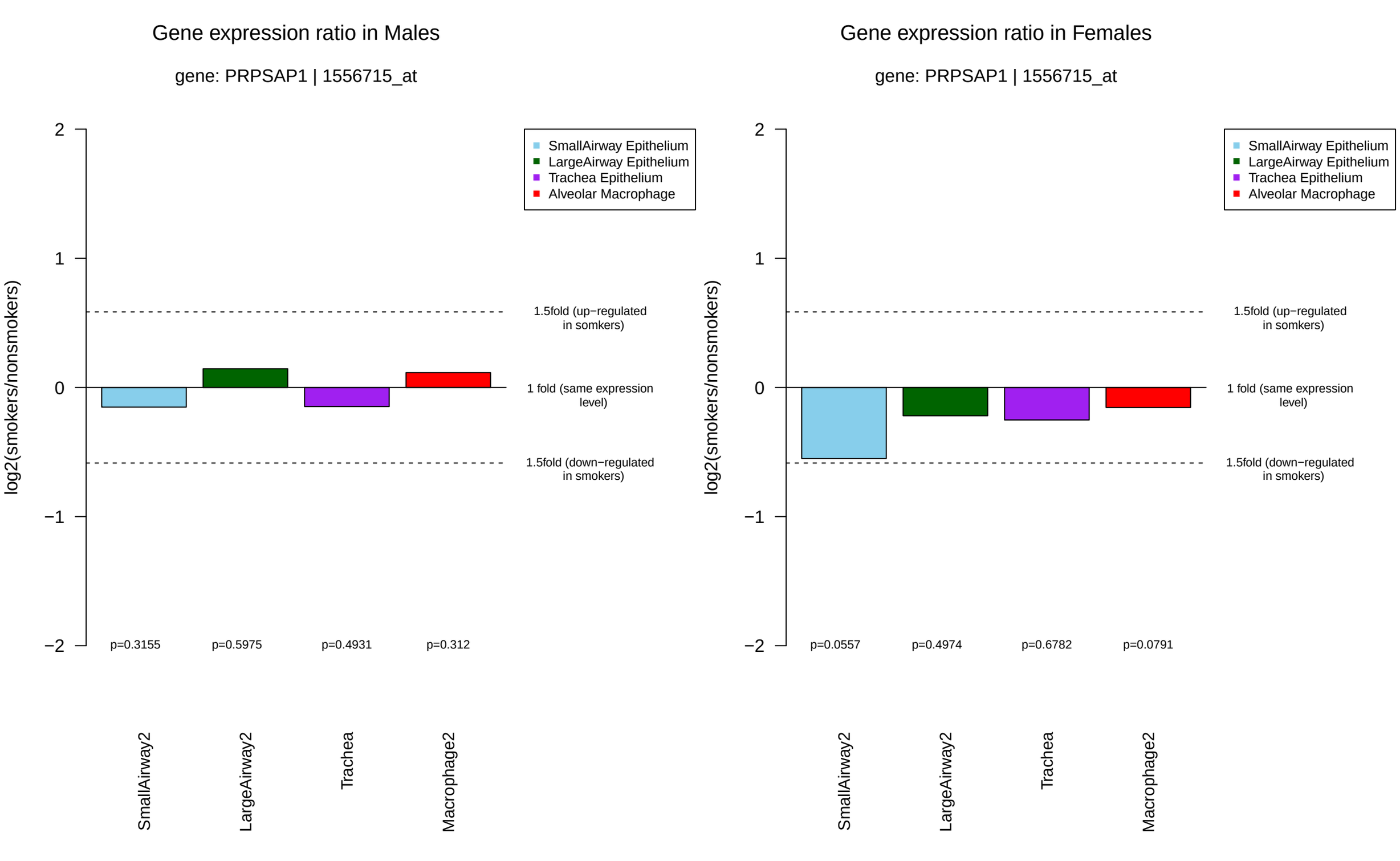
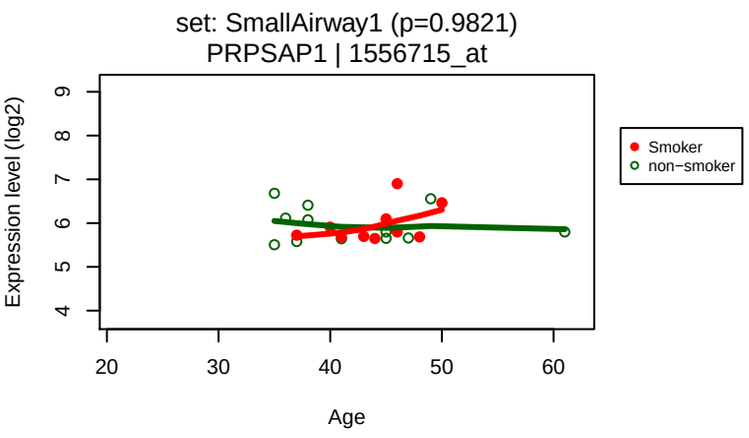
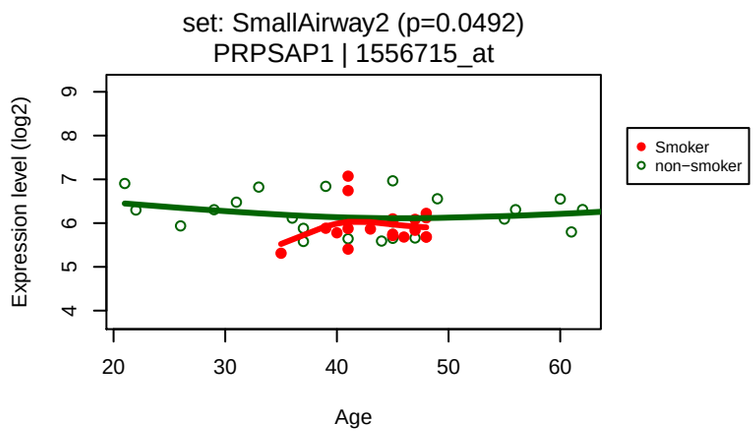
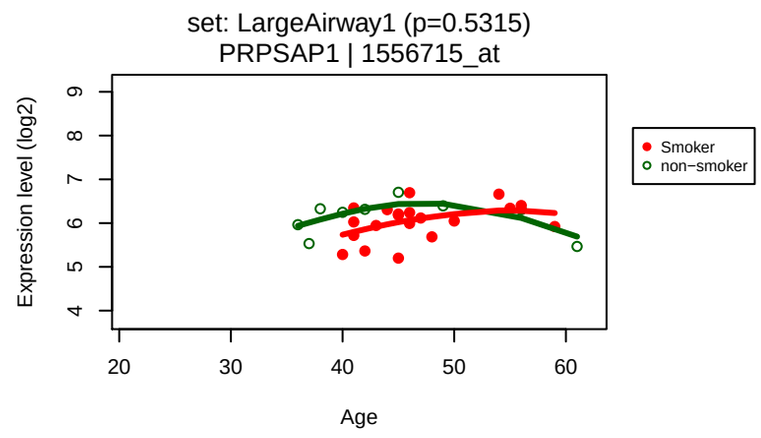

Gene expression of smokers and nonsmokers
gene: PRPSAP1 | 1556715_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
p=0.9821
p=0.0492
p=0.1571
p=0.5315
p=0.7752
p=0.2956
p=0.712
p=0.784
Smoker
non−smoker
Expression level (log2 transformed)
15
10
5
0
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

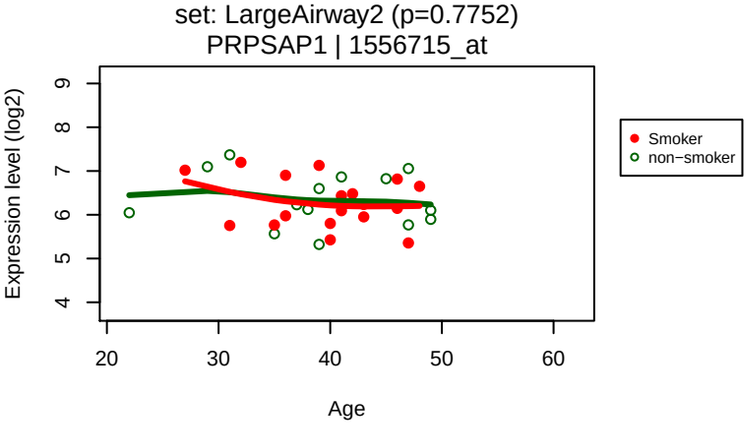

Gene expression ratio (smokers/nonsmokers)
gene: PRPSAP1 | 1556715_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Peripheral Lung Tissue
log2(smokers/nonsmokers)
2
1
0
−1
−2
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.9821
p=0.0492
p=0.1571
p=0.5315
p=0.7752
p=0.2956
p=0.712
p=0.784
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

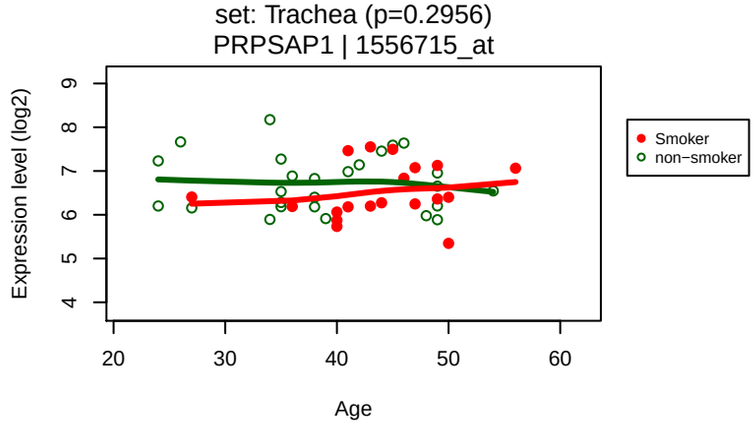

Gene expression ratio in Males
gene: PRPSAP1 | 1556715_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.3155
p=0.5975
p=0.4931
p=0.312
SmallAirway2
LargeAirway2
Trachea
Macrophage2
Gene expression ratio in Females
gene: PRPSAP1 | 1556715_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.0557
p=0.4974
p=0.6782
p=0.0791
SmallAirway2
LargeAirway2
Trachea
Macrophage2

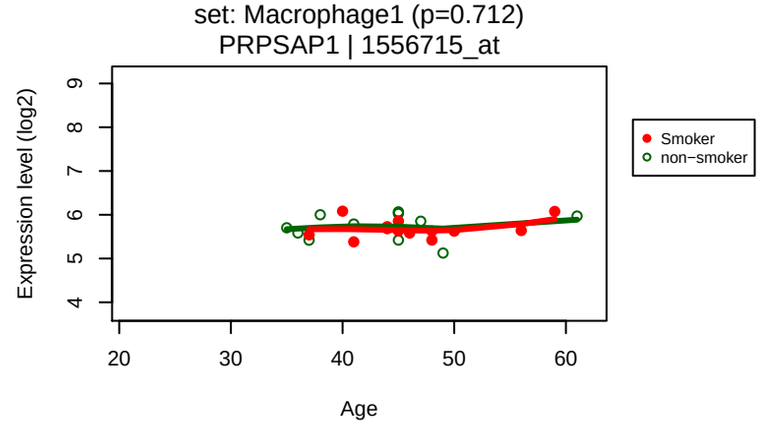

set: SmallAirway1 (p=0.9821)
PRPSAP1 | 1556715_at
Expression level (log2)
Age
Smoker
non−smoker

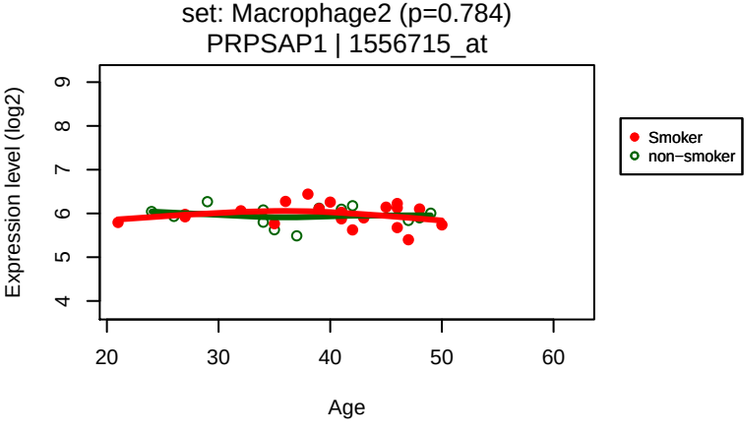

set: SmallAirway2 (p=0.0492)
PRPSAP1 | 1556715_at
Expression level (log2)
Age
Smoker
non−smoker

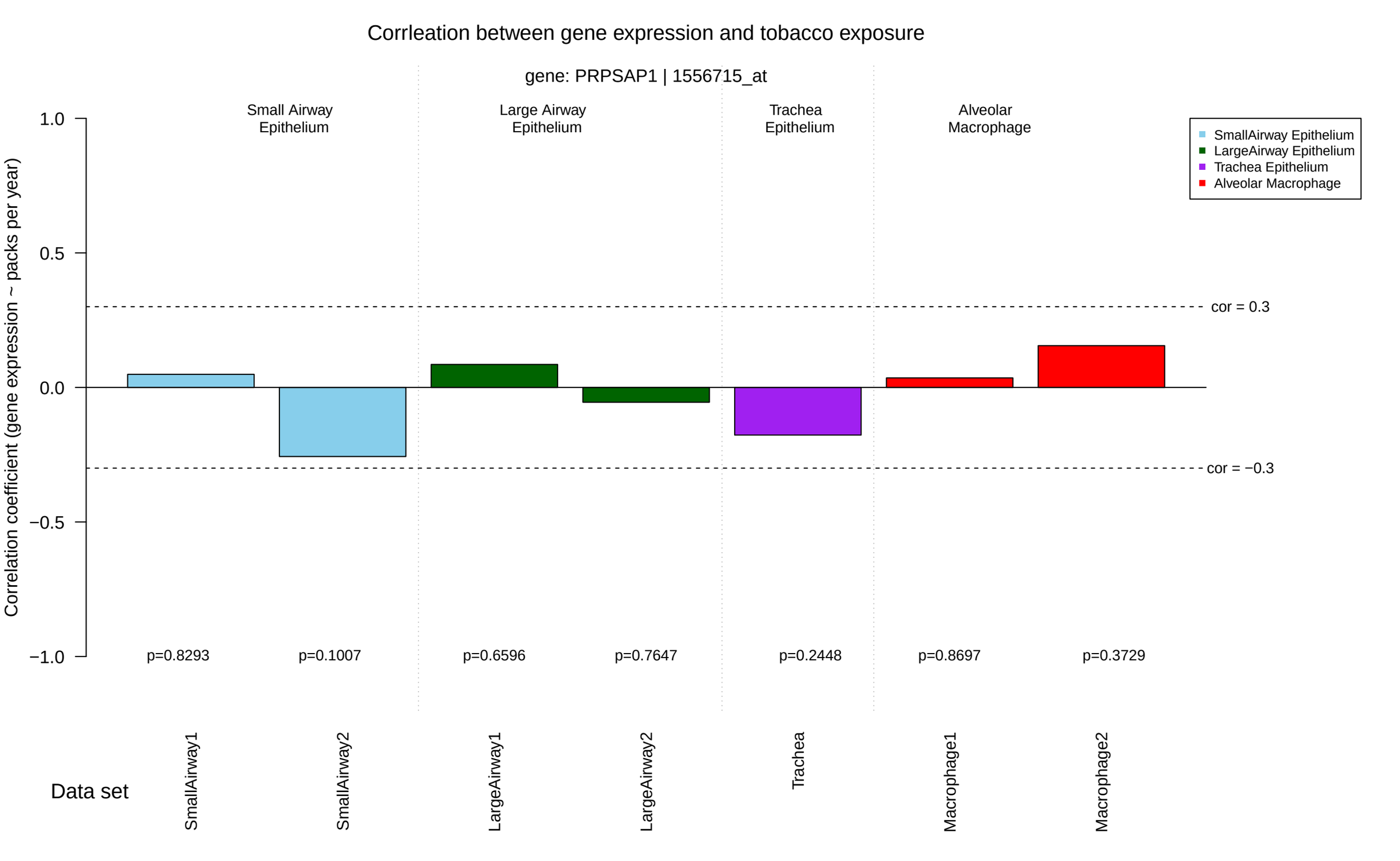

set: LargeAirway1 (p=0.5315)
PRPSAP1 | 1556715_at
Expression level (log2)
Age
Smoker
non−smoker


set: LargeAirway2 (p=0.7752)
PRPSAP1 | 1556715_at
Expression level (log2)
Age
Smoker
non−smoker


set: Trachea (p=0.2956)
PRPSAP1 | 1556715_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage1 (p=0.712)
PRPSAP1 | 1556715_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage2 (p=0.784)
PRPSAP1 | 1556715_at
Expression level (log2)
Age
Smoker
non−smoker


Corrleation between gene expression and tobacco exposure
gene: PRPSAP1 | 1556715_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Correlation coefficient (gene expression ~ packs per year)
1.0
0.5
0.0
−0.5
−1.0
cor = 0.3
cor = −0.3
p=0.8293
p=0.1007
p=0.6596
p=0.7647
p=0.2448
p=0.8697
p=0.3729
Data set
SmallAirway1
SmallAirway2
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2